IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

<Name>
<Date>

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies

This project analyzes SpaceX Falcon 9 launch data to identify key factors influencing launch success rates. Data was gathered from Wikipedia using **Beautiful Soup** and **Pandas**, and analyzed using **SQL**.

- Data Collection and transform to DataFrame
  - **Wikipedia:** Web scraping was performed using **Requests and Beautiful Soup** to extract launch records from HTML tables.
  - **SpaceX API:** requests.get() was used to retrieve JSON data, which was converted into a DataFrame using **Pandas json_normalize()**.
  - Information from both sources was combined into a dataset for further analysis.
- Perform Data Wrangling
- EDA (Explanatory Data Analysis) and SQL
  - **SQL** was used to query, aggregate, and analyze the launch dataset.
  - **Matplotlib and Seaborn** were used to visualize relationships between launch success and key variables.
- Interactive visual analysis
  - **Folium** was used to map launch sites and examine geographical patterns.
  - **Plotly Dash** was used to create interactive visualizations for **launch site and payload analysis**.
- Predictive analysis
  - Standardized numerical features using **StandardScaler** and split the data into **80% training and 20% test sets**.
  - Trained and compared **Logistic Regression, SVM, Decision Tree, and KNN** models.
  - Used **GridSearchCV with 10-fold cross-validation** to optimize hyperparameters.

Executive Summary

- Summary of all results

Launch success rates increased significantly from **2013 onward**, reaching around **80%** in recent years.
Different launch sites exhibited varying success rates, with some sites achieving higher payload capacities than others.
Certain orbit types, such as ES-L1 and GEO, had **100% success rates**, while others, like SO, showed a **0% success rate**.
Although high-payload launches were less frequent, they had a higher success proportion, indicating their reliability.
The Decision Tree model achieved the highest test accuracy of **94.4%** in predicting landing outcomes.
These insights contribute to a deeper understanding of the reliability and efficiency of reusable launches, guiding future improvements in SpaceX operations and informing strategic decisions for upcoming missions.

Introduction

- SpaceX has conducted numerous Falcon 9 launches, making the successful landing and reusing of boosters crucial for enhancing reusability and reducing launch costs. This innovative approach is essential for advancing space exploration and commercial spaceflight.

- However, the outcomes of launches and landings can vary significantly based on factors such as launch site, payload mass, and orbit type. Understanding these factors is vital for improving the reliability and efficiency of reusable launches, ultimately contributing to the sustainability of space missions.

- In this project, I analyzed historical Falcon 9 launch data to identify patterns related to launch and landing outcomes. Additionally, I built and compared various machine learning models to predict the success of booster landings, aiming to provide valuable insights for future launches.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - The project uses historical **SpaceX Falcon 9 launch data** originally collected from Wikipedia.
  - **Requests and BeautifulSoup** were used to scrape and extract launch records from Wikipedia HTML tables.
  - The extracted data was organized into a **Pandas DataFrame** for further analysis.
- Perform data wrangling
  - The dataset contains approximately **90 launch records**, including launch site, payload mass, orbit, booster version, and landing outcome.
  - The original Landing_Outcome variable contained multiple categories, including **True ASDS, True RTLS, False ASDS, and None None**.
  - These outcomes were transformed into a binary target variable, **Class**, for machine learning:
    - **Class = 1:** Successful first-stage landing
    - **Class = 0:** Unsuccessful first-stage landing

Methodology

Executive Summary

- Perform exploratory data analysis (EDA) using visualization and SQL

  Pandas was used for data manipulation and exploratory analysis.
  Matplotlib / Seaborn were used to visualize launch trends and relationships.
  Exploratory analysis examined relationships between launch success and **launch site, orbit, and payload mass**.

- Perform interactive visual analytics using Folium and Plotly Dash

  - **Folium** was used to map launch sites and visualize geographical patterns.
    Plotly Dash was used to create interactive visualizations for launch site and payload analysis.

Methodology

Executive Summary

- Perform predictive analysis using classification models
  - Classification models including **Logistic Regression, SVM, Decision Tree, and KNN** were trained to predict whether a Falcon 9 booster would successfully land.
  - Numerical features were standardized using **StandardScaler**, and the dataset was divided into **80% training and 20% test sets**. **GridSearchCV with 10-fold cross-validation** was used to optimize the model hyperparameters. The models were evaluated using their validation and test accuracy, with **Decision Tree achieving the highest test accuracy of 94.4%**.

Data Collection

- **Two data collection methods were used:**

SpaceX REST API: Launch Data → JSON Response → Pandas DataFrame

Wikipedia Web Scraping: HTML Tables → BeautifulSoup → Pandas DataFrame

SpaceX REST API: Used to obtain detailed launch information.
Wikipedia: Used to collect historical Falcon 9 launch records.
Pandas: Used to organize the collected data into DataFrames

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-spacex-data-collection-api.ipynb
https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-webscraping%20(1).ipynb

Data Collection – SpaceX API

SpaceX REST API

- **requests.get() – HTTP GET Request**

JSON Response

- **pd.json_normalize() – JSON → DataFrame**

Select Relevant Fields

- Rocket, payloads, launchpad, cores, flight_number, date_utc

Extract Launch Information

- rocket → Booster Version
- payloads → Payload Mass / Orbit
- launchpad → Launch Site / Latitude / Longitude
- cores → Landing Outcome / Flights / GridFins / Reused / Legs / Landing Pad

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

Data Collection - Scraping

HTTP GET Request from Wikipedia

- **requests.get()→HTML Response**

Parse HTML & Extract Launch Records

- **BeautifulSoup – BeautifulSoup()**

Find Target Tables

- **find_all("table")**

Extract Table Headers & Records

- find_all("th") / find_all("tr")

Extract Launch Information

- Date , Booster Version, Payload Mass, Launch Site, Landing Outcome

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-webscraping%20(1).ipynb

Data Wrangling

Filter Data

- Select Relevant Columns
- Remove launches with multiple cores
- Remove launches with multiple payloads

Extract & Transform Information

- rocket → Booster Version
- payloads → Payload Mass / Orbit
- launchpad → Launch Site / Latitude / Longitude
- cores → Landing Outcome / Flights / GridFins / Reused / Legs / Landing Pad

Handle Missing Values

- Payload Mass → replace missing values with the mean
- Landing Pad → retain None when no landing pad was used
- Date → remove records with missing dates

Create and Store the Final Dataset

The cleaned dataset was stored in the SQLite table **SPACEXTABLE** for SQL-based analysis.

https://github.com/HaruNNH/Final-Projects/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

- **Flight Number vs. Payload Mass – Scatter Plot**
  → Used to examine how payload mass and flight number are related to launch outcomes.
- **Flight Number vs. Launch Site – Scatter Plot**
  → Used to examine launch success patterns across different launch sites as flight numbers increased.
- **Payload Mass vs. Launch Site – Scatter Plot**
  → Used to examine the relationship between payload mass, launch site, and launch outcomes.
- **Orbit Type vs. Success Rate – Bar Chart**
  → Used to compare launch success rates across different orbit types.
- **Flight Number vs. Orbit Type – Scatter Plot**
  → Used to examine how orbit selection and launch outcomes changed as flight numbers increased.
- **Payload Mass vs. Orbit Type – Scatter Plot**
  → Used to examine the relationship between payload mass, orbit type, and launch outcomes.
- **Year vs. Success Rate – Line Chart**
  → Used to examine the yearly trend in launch success rates.

https://github.com/HaruNNH/Final-Projects/blob/main/edadataviz.ipynb

EDA with SQL

- Identified **unique launch sites** and filtered launch sites beginning with **"CCA"**.
- Calculated the **total payload mass** carried for NASA (CRS) missions.
- Calculated the **average payload mass** for the **F9 v1.1** booster.
- Identified the **date of the first successful ground-pad landing**.
- Identified boosters that achieved **successful drone-ship landings** with payloads between **4,000 and 6,000 kg**.
- Counted **successful and failed mission outcomes**.
- Identified booster versions that carried the **maximum payload mass** using a subquery.
- Retrieved **failed drone-ship landings in 2015**, including their dates, booster versions, and launch sites.
- Ranked **landing outcomes by frequency** between June 2010 and March 2017.

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

Build an Interactive Map with Folium

Map Objects and Visualization

Circles indicate the locations of the four SpaceX launch sites when the map is zoomed out.

When users **zoom in** on a launch site, individual launches are displayed as markers.

Green markers represent successful launches, while **red markers** represent unsuccessful launches.

Polylines connect CCAFS SLC-40 with nearby locations, including Titusville, Florida East Coast Railway, and Cheney Highway.

Distance labels show the distances between CCAFS SLC-40 and these locations, as well as the nearest coastline.

Interactive Features

The map dynamically provides more detailed information as users **zoom in** on a launch site.

The **latitude and longitude of the mouse pointer** are displayed in the upper-right corner.

Users can visually examine the distribution of successful and unsuccessful launches at each site.

Purpose

The circles provide a simple overview of the **locations of the launch sites** when zoomed out.

The individual markers allow users to examine the **success or failure of each launch** in greater detail.

The color coding makes it easy to identify successful and unsuccessful launches at a glance.

The distance measurements help visualize the **geographical relationship between CCAFS SLC-40 and nearby locations**.

https://github.com/HaruNNH/Final-Projects/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

Interactive Visualizations

An interactive dashboard was created using Plotly Dash.

The dashboard includes a pie chart showing the success rate for each launch site and a scatter plot showing the relationship between payload mass, booster version, and launch outcome.

Interactions

A dropdown menu allows users to select one of the four launch sites or ALL.

Based on the selected launch site, the pie chart dynamically displays its success rate.

When ALL is selected, the pie chart shows the proportion of total successful launches contributed by each launch site.

A range slider allows users to select a desired range of payload mass for the scatter plot.

The scatter plot displays launch outcomes within the selected payload range, with points color-coded by booster version.

Purpose

The interactive pie chart makes it easy to compare launch success across different launch sites.

The scatter plot helps visualize how payload mass and booster version are associated with launch outcomes.

These interactive features allow users to explore the data from different perspectives and identify patterns more easily.

https://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py

Predictive Analysis (Classification)

Data Preprocessing
- Class was used as the target variable.
- Features were standardized using StandardScaler.
- The data was split into training and test sets (80% / 20%).

Model Training
- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbors (KNN)

Hyperparameter Tuning
- GridSearchCV with 10-fold cross-validation was used to select the optimal hyperparamerset

Model Evaluation
- Test accuracy
- Confusion matrix

Model Comparison
- The performance of the four classification models was compared using test-set accuracy

http://github.com/HaruNNH/Final-Projects/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20(1).ipynb

Results

- Launch success rates increased substantially over time, particularly from 2013 onward, reaching around 80% in recent years.
- Launch sites showed different success patterns, with KSC LC-39A recording the highest success rate.
- Higher payload masses were not necessarily associated with lower success rates, and successful launches were observed across a wide range of payload masses.
- ES-L1, GEO, HEO, and SSO recorded 100% success rates, while SO recorded the lowest success rate at 0%.
- As Flight Number increased, additional orbit types such as SSO and HEO appeared, while GEO was selected only after Flight Number 80.
- The Decision Tree was the best-performing classification model, achieving the highest test accuracy of 94.4%.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

Method

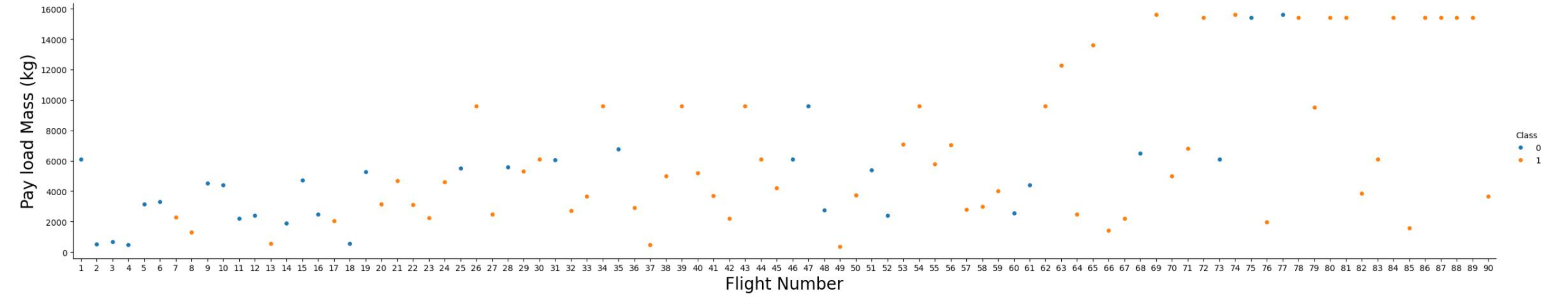
I mainly used **matplotlib** and **Seaborn** to analyze the collected data and visualize the results in graphs

Purpose:

The scatter plot shows the relationship between payload mass and flight number, with different colors indicating whether the launch was successful or unsuccessful. Examining changes in payload mass and launch success over time.

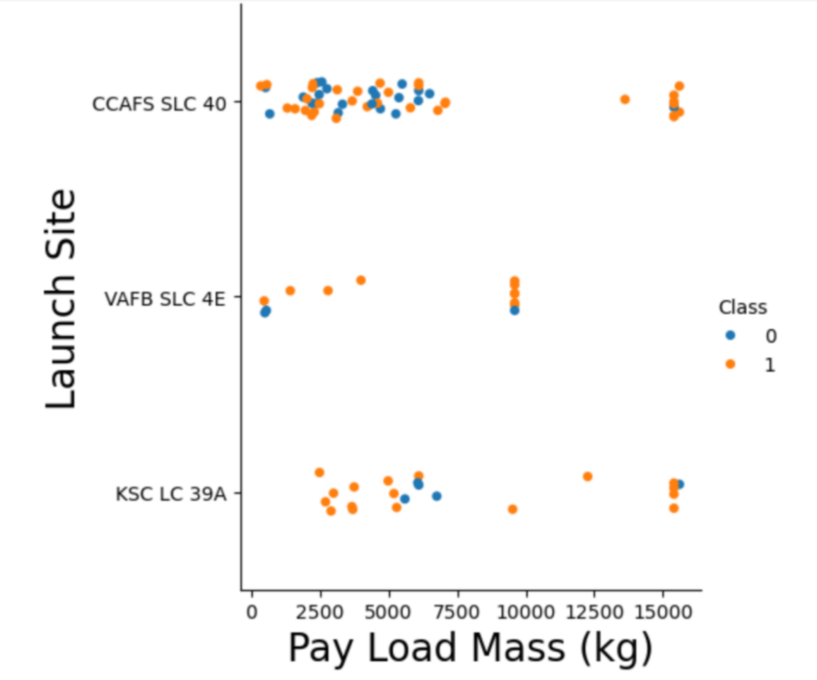
Findings:

As the flight number increases, the payload mass generally becomes heavier. Payloads exceeding 1,600 kg first appear around Flight 70. At the same time, the launch success rate shows an increasing trend over time.



https://github.com/HaruNNH/Final-Projects/blob/main/edadataviz.ipynb

Payload vs. Launch Site



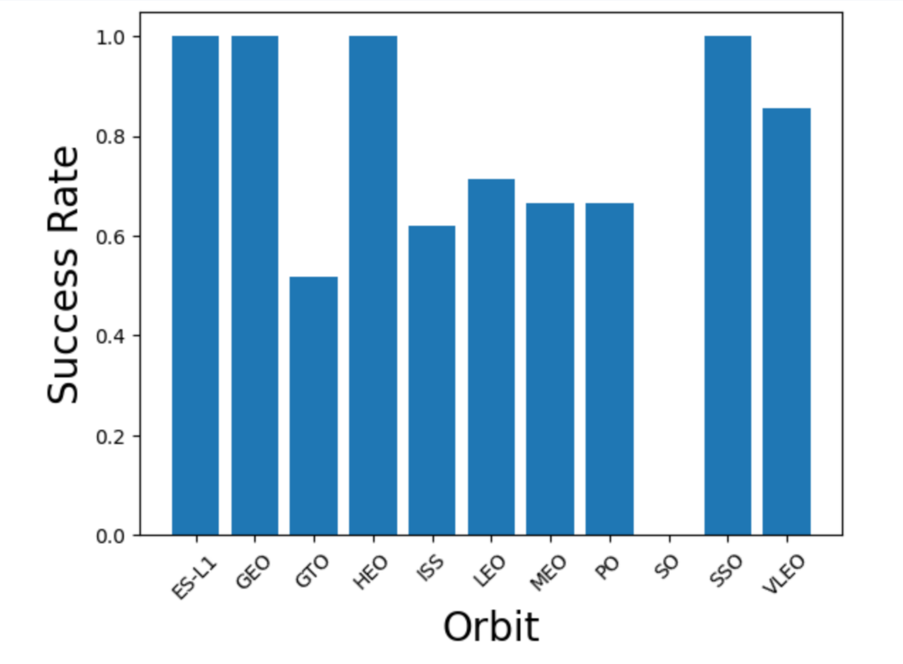
Purpose: The scatter plot illustrates the relationship between payload mass, launch site, and launch outcome (success or failure). This visualization aims to reveal how payload weight and launch site influence launch success rates.

Findings

- Smaller payloads, particularly those weighing less than 7,500 kg, were more common. However, heavier payloads appear to have a higher success rate.
- At **VAFB SLC 4E**, there were no launches with payloads exceeding 10,000 kg.
- **CCAFS SLC 40** had the highest number of launches, but it also had the highest number of failed launches among the launch sites.
- At **KSC LC 39**, launches with payloads below 2,500 kg were relatively rare.

https://github.com/HaruNNH/Final-Projects/blob/main/edadataviz.ipynb

Success Rate vs. Orbit Type



Purpose

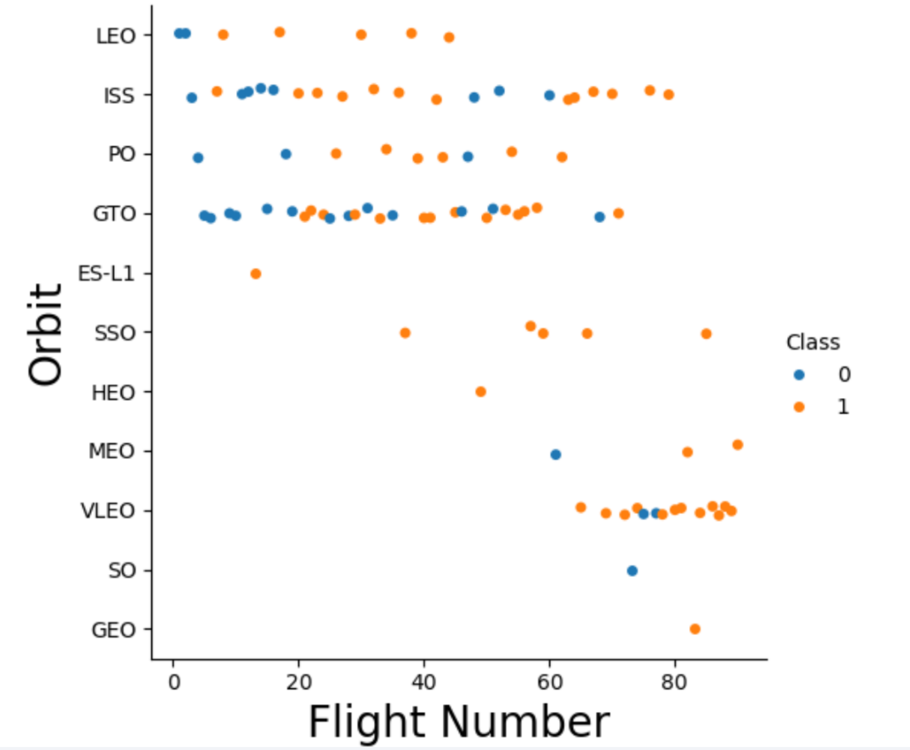
The following bar graph illustrates the relationship between orbit type and success rate. This visualization aims to identify the most appropriate orbit for launch based on the success rates recorded for each orbit, which can inform future launch strategies.

Findings

- The orbits with the highest success rates—ES-L1, GEO, HEO, and SSO—each recorded a 100% success rate, indicating their reliability for successful launches.
- Following these, VLEO achieved an 80% success rate.
- LEO has a success rate of approximately 70%, while MEO and PO have similar rates of slightly over 60%. ISS recorded a 60% success rate, and GTO had a success rate of around 50%.
- Notably, SO had the lowest success rate at 0%, highlighting potential challenges associated with this orbit.

https://github.com/HaruNNH/Final-Projects/blob/main/edadataviz.ipynb

Flight Number vs. Orbit Type



Purpose

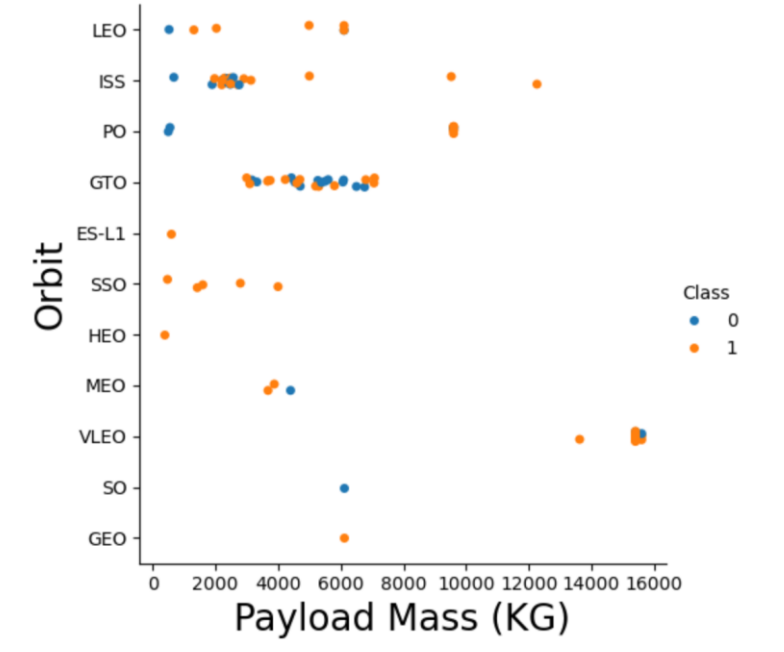
The following chart illustrates the relationship between **flight number and orbit type**. This visualization aims to examine how **orbit selection changed as the flight number increased** and identify trends in orbit selection over time.

Findings

- Up to **flight number 40**, launches mainly used **LEO, ISS, PO, GTO, and ES-L1**. ES-L1, however, was selected only once.
- After **flight number 40**, **SSO and HEO** began to appear among the selected orbits.
- **MEO, VLEO, and SO** first appeared after **flight number 60**, while **GEO** was selected for the first time after **flight number 80**.
- **ISS** had the widest range of use and was selected multiple times between **flight numbers 0 and 80**.
- After **flight number 60**, **VLEO** was the most frequently selected orbit.

https://github.com/HaruNNH/Final-Projects/blob/main/edadataviz.ipynb

Payload vs. Orbit Type



Purpose

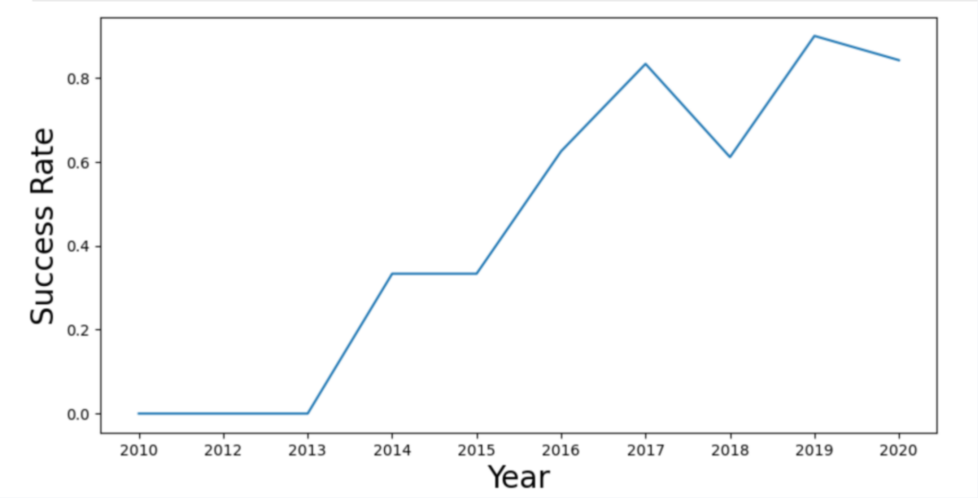
The following scatter plot illustrates the relationship between **payload mass, orbit, and launch outcome (success or failure)**. It aims to show how payload mass and orbit selection affect launch success.

Findings

- **GEO, HEO, and ES-L1** each have only one launch attempt, making clear patterns difficult to identify.
- **GTO** has many attempts and a relatively high failure rate, with payloads mainly between **4,000 and 8,000 kg**.
- **LEO, PO, and ISS** show failures mainly with payloads below **4,000 kg**, while heavier payloads have no observed failures. **SSO** has several attempts, all successful.
- **VLEO** sometimes carries the heaviest payloads, reaching about **16,000 kg**, with successful launches more common than failures.
- Overall, **heavier payloads tend to have higher success rates, while orbit selection also affects launch outcomes.**

https://github.com/HaruNNH/Final-Projects/blob/main/edadataviz.ipynb

Launch Success Yearly Trend



Purpose

The graph shows changes in launch success rates over time, highlighting the improvement in SpaceX's launch reliability. This analysis helps identify trends in performance and factors related to launch success.

Findings

The success rate generally increased from 2013 to 2019, with a significant rise of about 30 percentage points from 2013 to 2014. The rate remained stable in 2015, then increased to around 60% in 2016. Notably, it first reached **80% in 2017**, but experienced a drop to 60% in 2018 before reaching its highest success rate of **over 80% in 2019**.

This fluctuation highlights the challenges faced in 2018, which may warrant further investigation into factors that affected launch outcomes that year.
Overall, the trend indicates a positive trajectory in SpaceX's launch success rates over time.

https://github.com/HaruNNH/Final-Projects/blob/main/edadataviz.ipynb

All Launch Site Names

Display the names of the unique launch sites in the space mission

SQL example:
pd.read_sql("SELECT DISTINCT Launch_Site FROM SPACEXTABLE", con)

Result:
There are four different sites, CCAFS LC-40, VAFB SLC-45, KSC LC-39A, and CCAFS SLC-40.

	Launch_Site
0	CCAFS LC-40
1	VAFB SLC-4E
2	KSC LC-39A
3	CCAFS SLC-40

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

Launch Site Names Begin with 'CCA'

Display launches whose launch sites begin with the string 'CCA'

SQL example:
pd.read_sql("SELECT *FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5", con)

Result:
The five missions shown below were all successful. However, **two landing attempts failed, while three missions did not attempt a landing at all**. All five missions shared the same **LEO orbit** and **F9 V1.0 booster version**.

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
0	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
1	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of...	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
3	2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
4	2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

Total Payload Mass

Purpose: Display the total payload mass carried by boosters launched by NASA (CRS)

SQL example:
pd.read_sql("SELECT SUM(PAYLOAD_MASS__KG_) AS total_payload_mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)'", con)

Result:
It was found that the **total payload mass launched by NASA was 45596 kg**.

	total_payload_mass
0	45596

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

Average Payload Mass by F9 v1.1

Display average payload mass carried by booster version F9 v1.1

SQL example:
pd.read_sql("SELECT AVG(PAYLOAD_MASS__KG_) AS average_payload_mass FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1'", con)

Result:
It was found that the **average payload mass carried by F9 V1.1 boosters was 2,928.4 kg**.

	average_payload_mass
0	2928.4

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

First Successful Ground Landing Date

List the date when the first succesful landing outcome in ground pad was achieved.

SQL example:
pd.read_sql("SELECT MIN(Date) AS first_successful_ground_landing FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)'", con)

Result:
The first successful ground landing was on December 22, 2015.

	first_successful_ground_landing
0	2015-12-22

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

Successful Drone Ship Landing with Payload between 4000 and 6000

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000.

SQL example:
pd.read_sql("SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ between 4000 and 6000", con)

Result:
There are four booster version F9 FT B1022, F9 FT B1026, F9 FT B1021.2, and F9 FT B1031.2

	Booster_Version
0	F9 FT B1022
1	F9 FT B1026
2	F9 FT B1021.2
3	F9 FT B1031.2

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

SQL example:

```
pd.read_sql("""
SELECT "Mission_Outcome", COUNT(*) AS count
FROM SPACEXTABLE GROUP BY "Mission_Outcome"""", con)
```

Result:

Almost all missions were successful, with only **one in-flight failure** recorded in total.

	Mission_Outcome	count
0	Failure (in flight)	1
1	Success	98
2	Success	1
3	Success (payload status unclear)	1

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

Boosters Carried Maximum Payload

List all the booster_versions that have carried the maximum payload mass

SQL example:

```
pd.read_sql("""
SELECT "Booster_Version"
FROM SPACEXTABLE
WHERE "PAYLOAD_MASS__KG_" = (
   SELECT MAX("PAYLOAD_MASS__KG_")
   FROM SPACEXTABLE
)""", con)
```

Result:

There are 12 booster versions which carried maximum payload mass

	Booster_Version
0	F9 B5 B1048.4
1	F9 B5 B1049.4
2	F9 B5 B1051.3
3	F9 B5 B1056.4
4	F9 B5 B1048.5
5	F9 B5 B1051.4
6	F9 B5 B1049.5
7	F9 B5 B1060.2
8	F9 B5 B1058.3
9	F9 B5 B1051.6
10	F9 B5 B1060.3
11	F9 B5 B1049.7

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

2015 Launch Records

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

SQL example:

pd.read_sql("""SELECT Date, "Landing_Outcome", "Booster_Version", "Launch_Site"
FROM SPACEXTABLE WHERE Date LIKE '2015%'
  AND "Landing_Outcome" = 'Failure (drone ship)' """, con)

Result:

Among the 2015 missions, **two in-drone-ship landing attempts ended in failure**, on **January 10 and April 14**, respectively. Both missions used the **F9 V1.1 booster** and were launched from **CCAFS LC-40**

	Date	Landing_Outcome	Booster_Version	Launch_Site
0	2015-01-10	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
1	2015-04-14	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

SQL example:

```
pd.read_sql("""
SELECT "Landing_Outcome", COUNT(*) AS count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY count DESC""", con)
```

Result:

No attempt was the most frequent outcome, with a count of 10, followed by Success (drone ship) and Failure (drone ship), with 5 each.

	Landing_Outcome	count
0	No attempt	10
1	Success (drone ship)	5
2	Failure (drone ship)	5
3	Success (ground pad)	3
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Failure (parachute)	2
7	Precluded (drone ship)	1

https://github.com/HaruNNH/Final-Projects/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

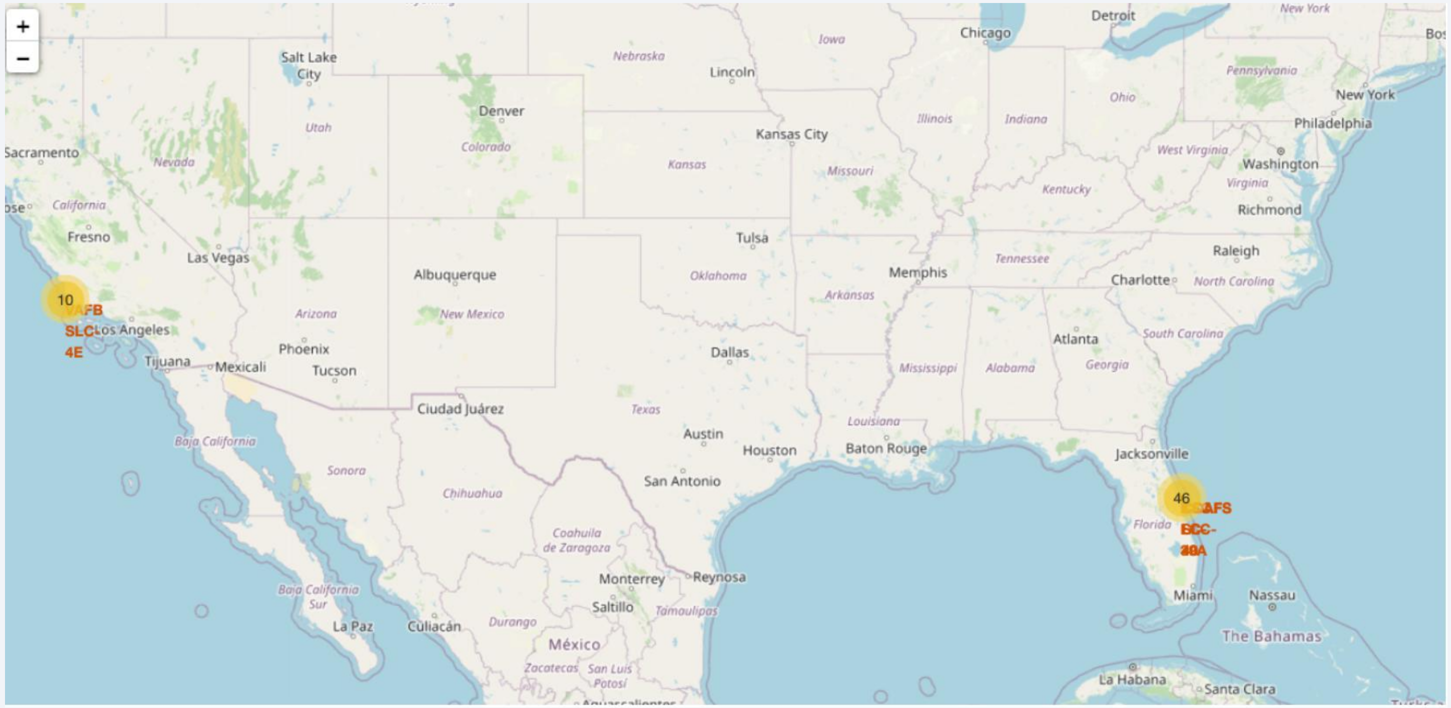
Section 3

Launch Sites Proximities Analysis

Locations of the Four Launch Sites

The following map was created using Folium. It contains 56 launch locations, with each record showing the launch site, geographic coordinates, and launch success class. Most launches took place at sites on the East Coast, with 46 launches.

VAFB SLC-4E is located on the West Coast, while the others—CCAFS LC-40, CCAFS SLC-40, and KSC LC-39A—are located on the East Coast.

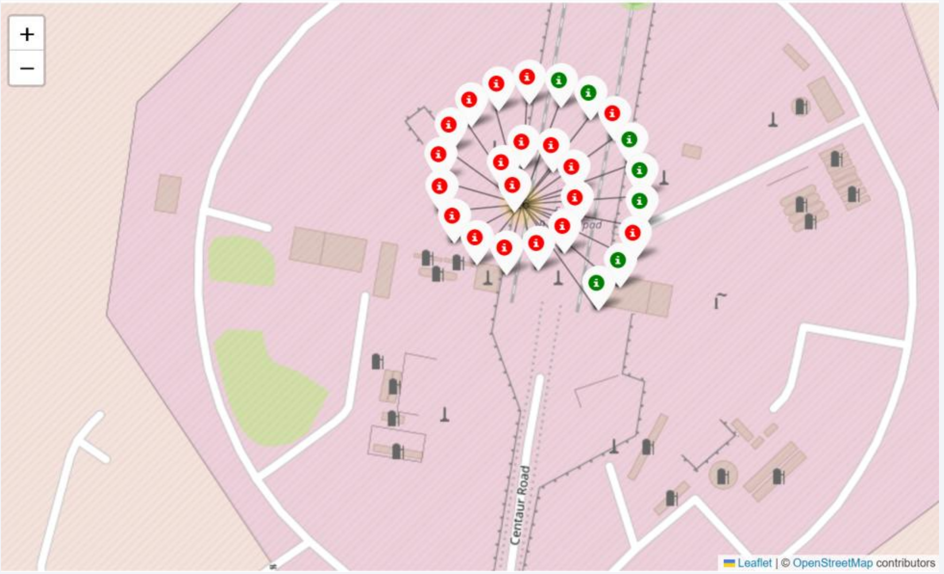


https://github.com/HaruNNH/Final-Projects/blob/main/lab_jupyter_launch_site_location.ipynb

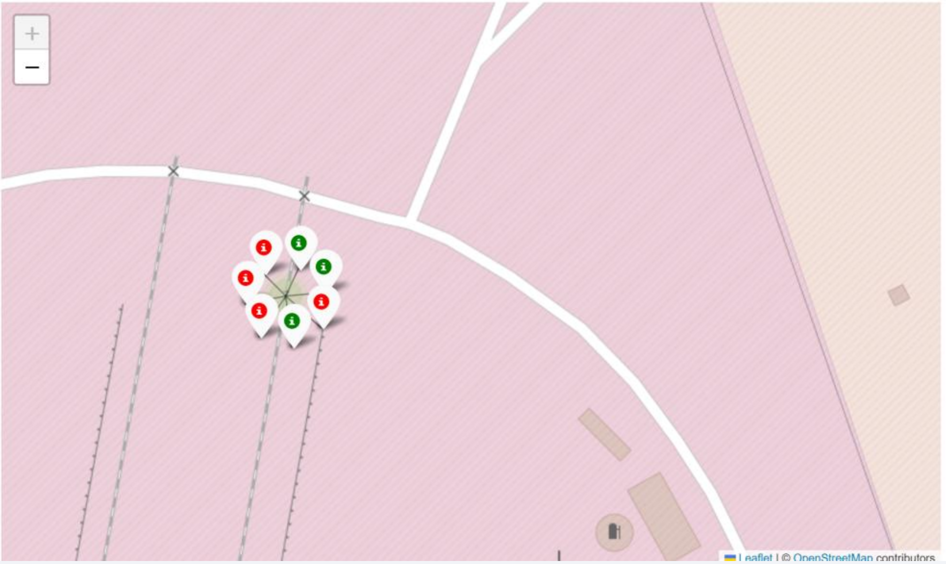
Launch Success Visualization

The following two maps show the success and failure of launches from CCAFS LC-40 and CCAFS SLC-40.
Green markers indicate successful launches, while red markers indicate failures.
There were 26 launches from CCAFS LC-40 and 7 launches from CCAFS SLC-40, showing that the number of launches varied significantly between the two sites.
As shown here, **launches from CCAFS LC-40 are dominated by red markers, indicating that the number of failures significantly exceeds the number of successful launches.**

CCAFS LC-40



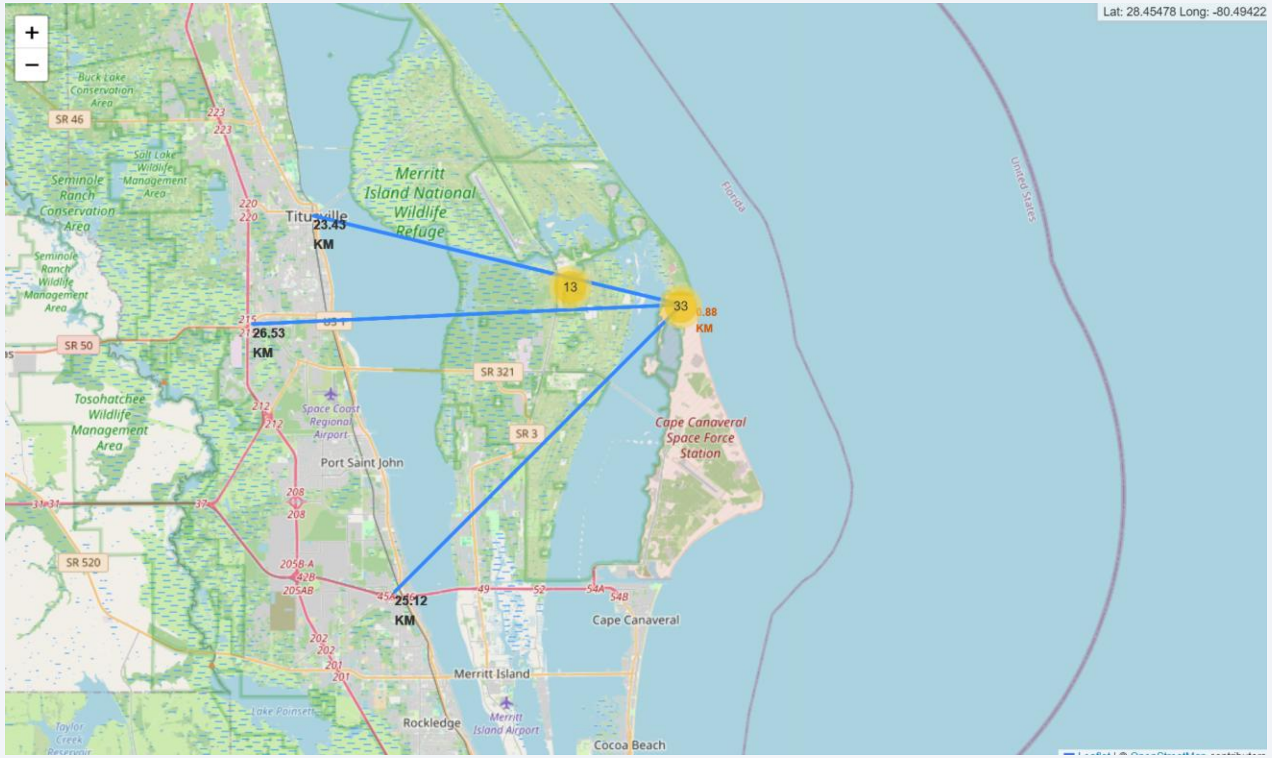
CCAFS SLC-40



https://github.com/HaruNNH/Final-Projects/blob/main/lab_jupyter_launch_site_location.ipynb

Distance Between Two Locations

Using **Folium**, I calculated the distances between the CCAFS SLC-40 launch site and the nearest **city, railway, and highway**. The distance to **Titusville** was **24.43 km**, while the distances to the **Florida East Coast Railway** and **Cheney Highway** were **25.12 km** and **26.53 km**, respectively.
The distance to the **nearest coastline** was **0.88 km**, indicating that **CCAFS SLC-40 is located relatively close to the coast**.



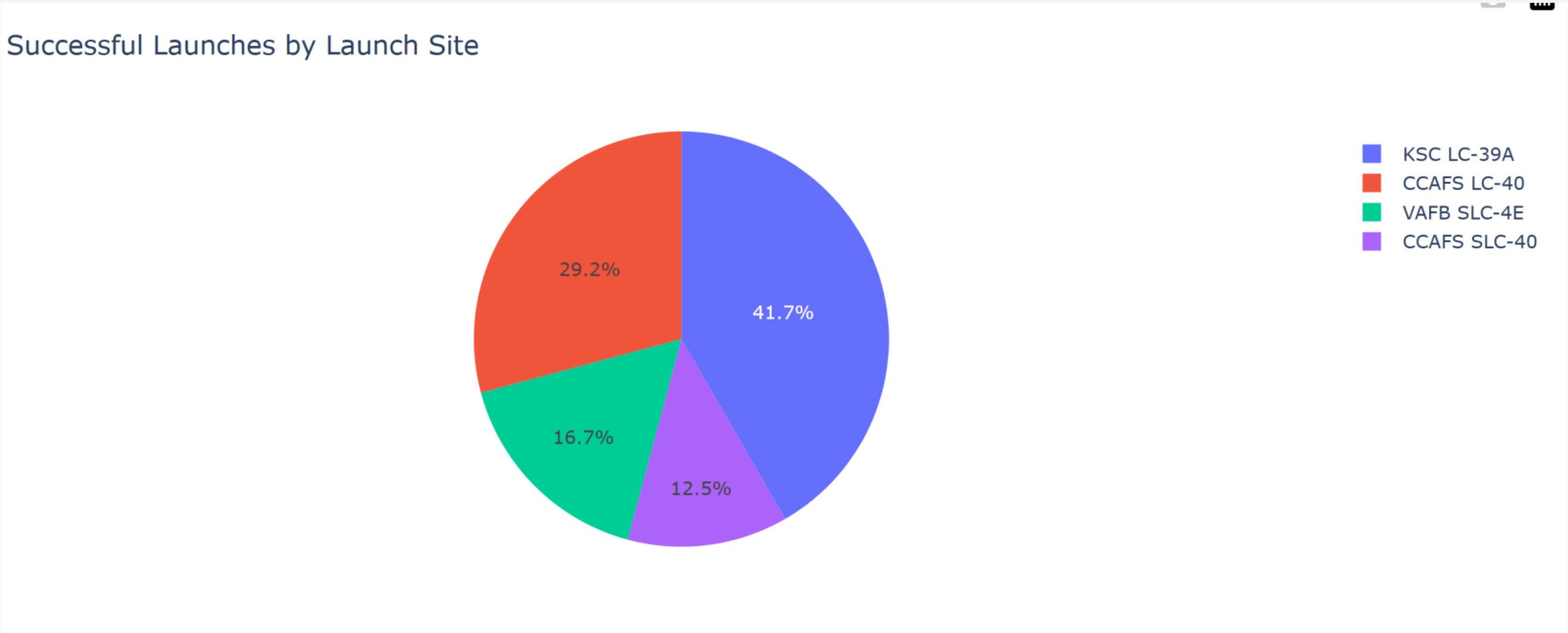
https://github.com/HaruNNH/Final-Projects/blob/main/lab_jupyter_launch_site_location.ipynb

Section 4

Build a Dashboard with Plotly Dash

Pie Chart of All Launch Sites

I used Plotly Dash to create an interactive graph.
This pie chart shows the proportion of all successful launches that were carried out from each specific launch site.



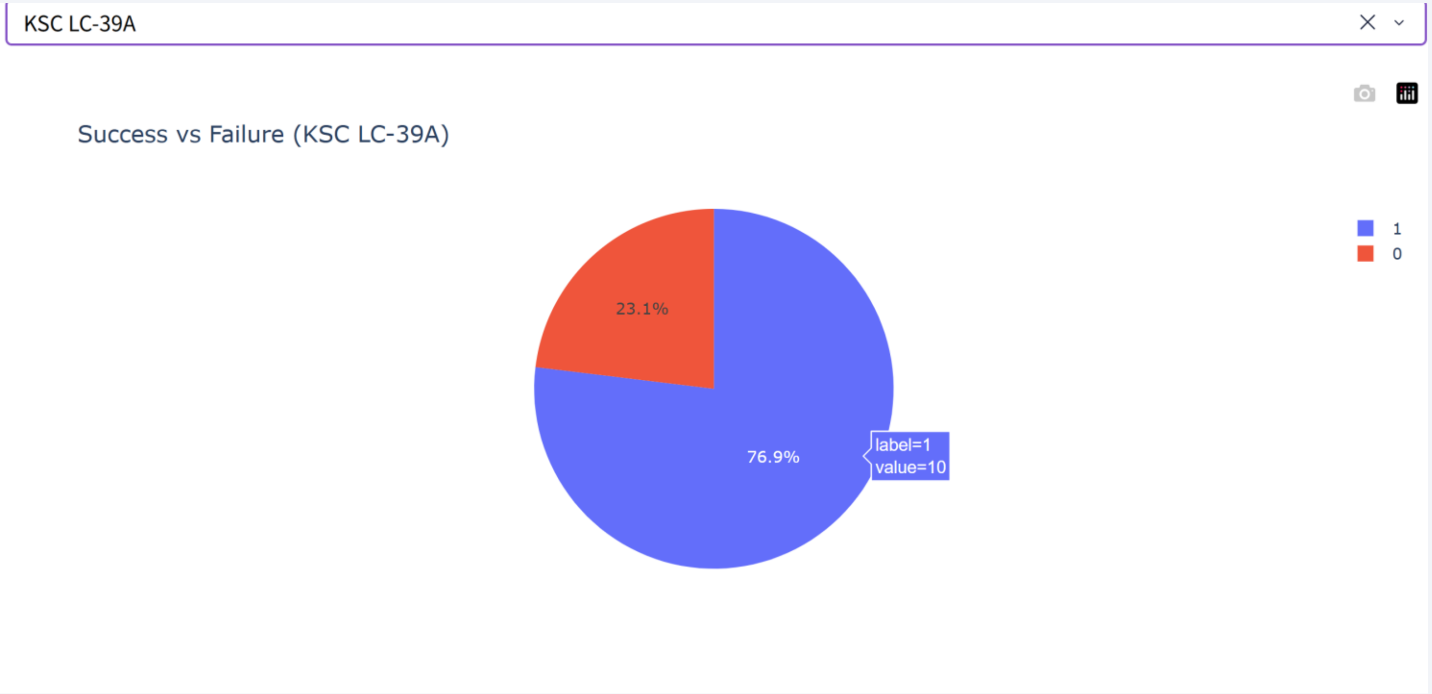
KSC LC-39A recorded the highest number of successful launches, accounting for 41.7% of all successful launches. This was followed by CCAFS LC-40 at 29.2%, VAFB SLC-4E at 16.7%, and CCAFS SLC-40 at 12.5%.

https://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py

Pie Chart of Success rate on KSC LC-39A

This pie chart shows the success rate at **KSC LC-39A**, which recorded the **highest success rate among all launch sites**.

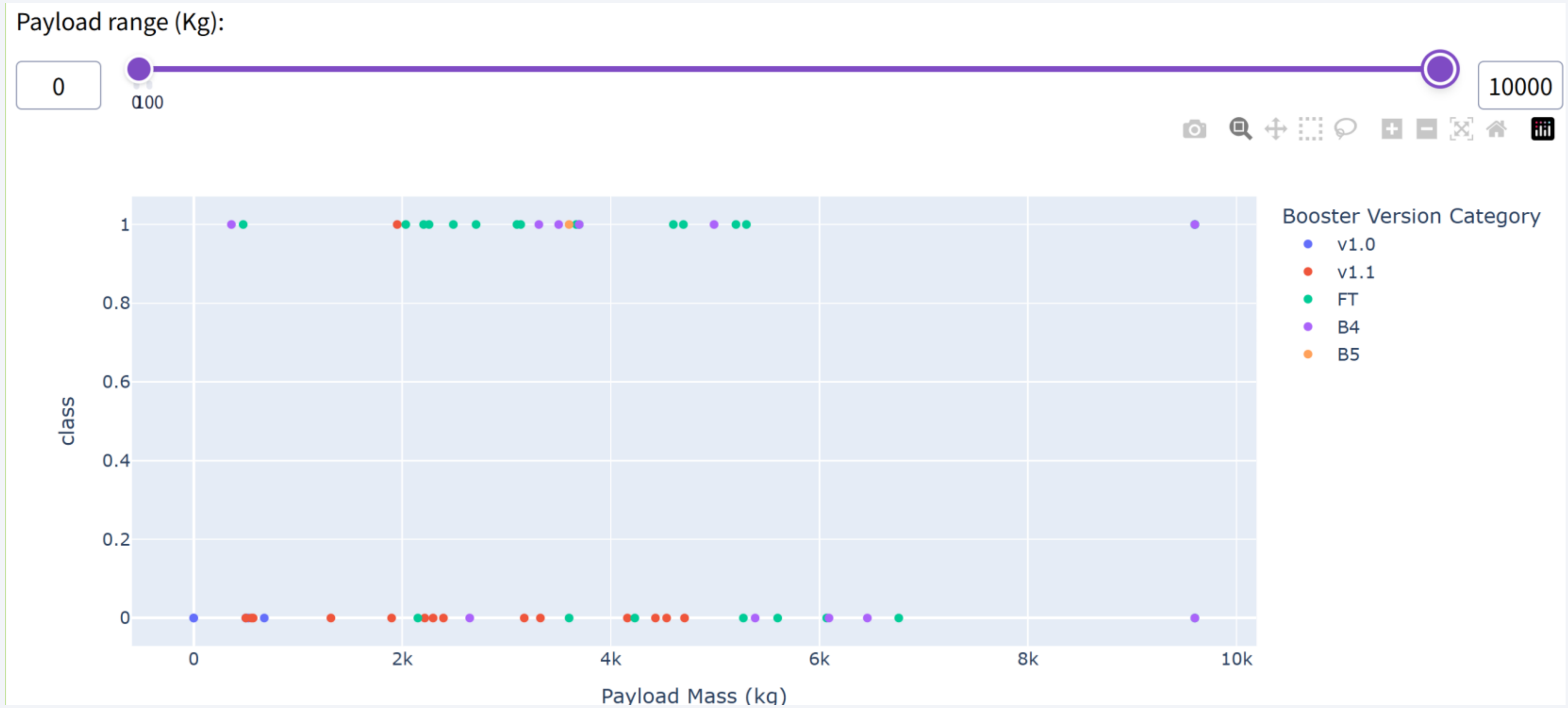
A value of 1 represents a successful launch.
10 successful launches were recorded, resulting in a success rate of 76.9%



https://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py

Relationship between Payload Mass and Booster

This scatter plot includes launches from all launch sites and shows the relationship between **payload mass and launch success**, with points color-coded by **booster version**. This allows the relationship between these two factors to be visualized. **FT and B4 account for the highest number of successful launches, while failures are more common with V1.1.** Launches carrying more than **8,000 kg** used the **B4 booster**, while **V1.0, V1.1, and B5** were not used for launches with payloads exceeding **6,000 kg**.

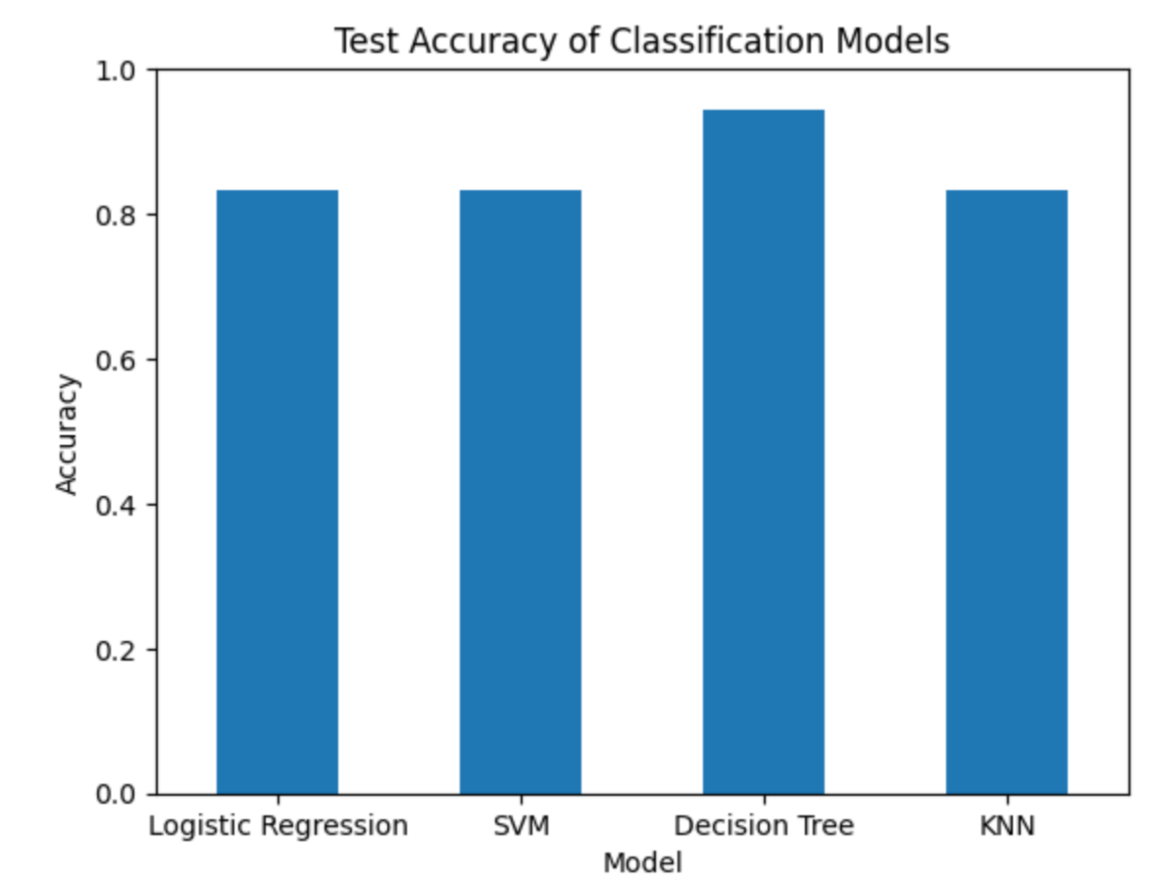


https://github.com/HaruNNH/Final-Projects/blob/main/spacex-dash-app.py

Section 5

Predictive Analysis (Classification)

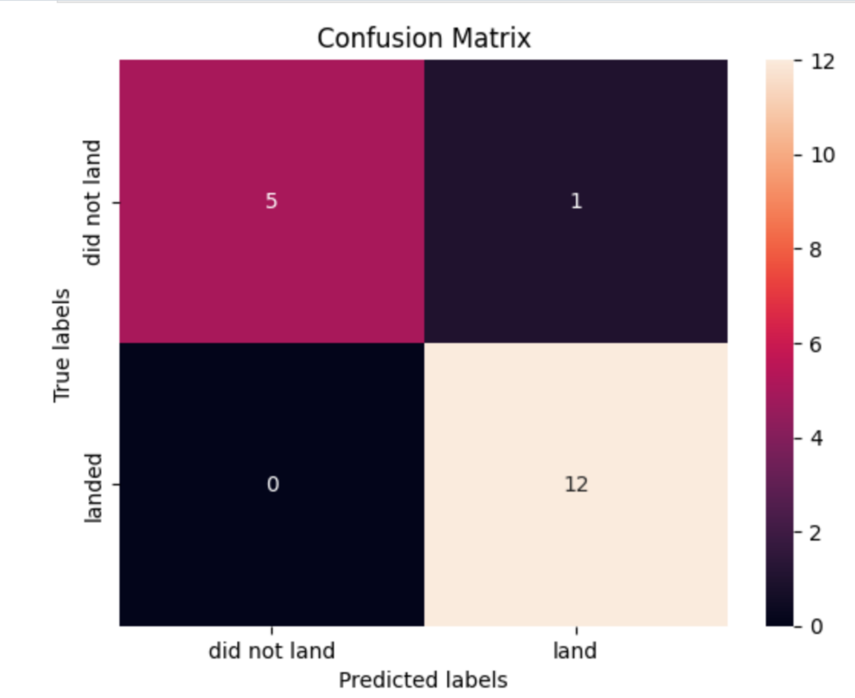
Classification Accuracy



The figure above shows the **test accuracy of the four classification models: Logistic Regression, SVM, Decision Tree, and KNN** as a bar chart.
The **Decision Tree** model has a higher test accuracy than the other models, exceeding **0.9**, while the other models achieve only slightly above **0.8**.

http://github.com/HaruNNH/Final-Projects/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20(1).ipynb

Confusion Matrix



Tree Decision— Results

Best parameters: criterion = gini, max_depth = 4, max_features = sqrt, min_samples_leaf = 2, min_samples_split = 10, splitter = random
Validation accuracy: 87.5%
Test accuracy: 94.4%

For the **18 test predictions**:

- **13 launches were predicted to land**. Of these, **12 actually landed**, while **1 did not**.
- **5 launches were predicted not to land**, and **all 5 actually did not land**.

This suggests that the Decision Tree model performed better than the other models for predicting landing success.

Overall, the Decision Tree performed best on the test data.

http://github.com/HaruNNH/Final-Projects/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5%20(1).ipynb

Conclusions

- The analysis indicates that Falcon 9 launch success is influenced by several key factors, including launch site, orbit, payload mass, and booster version.

- Exploratory and geographical analyses revealed significant differences in launch success rates across various sites and booster versions, highlighting the importance of these characteristics in operational strategies.

- Furthermore, machine learning analysis demonstrated that the Decision Tree model outperformed other models, achieving an impressive accuracy of **94.4%** on the test data.

- Overall, this analysis suggests that both launch characteristics and booster technology play critical roles in predicting Falcon 9 landing success. However, it is important to note that the dataset is relatively small, with few records, so the results should be interpreted with caution. Future work should aim to incorporate a larger dataset and explore additional variables that might further enhance predictive capabilities.

Appendix

- All GitHub URLs for this project are compiled in one place. Please access them here.

Thank you!